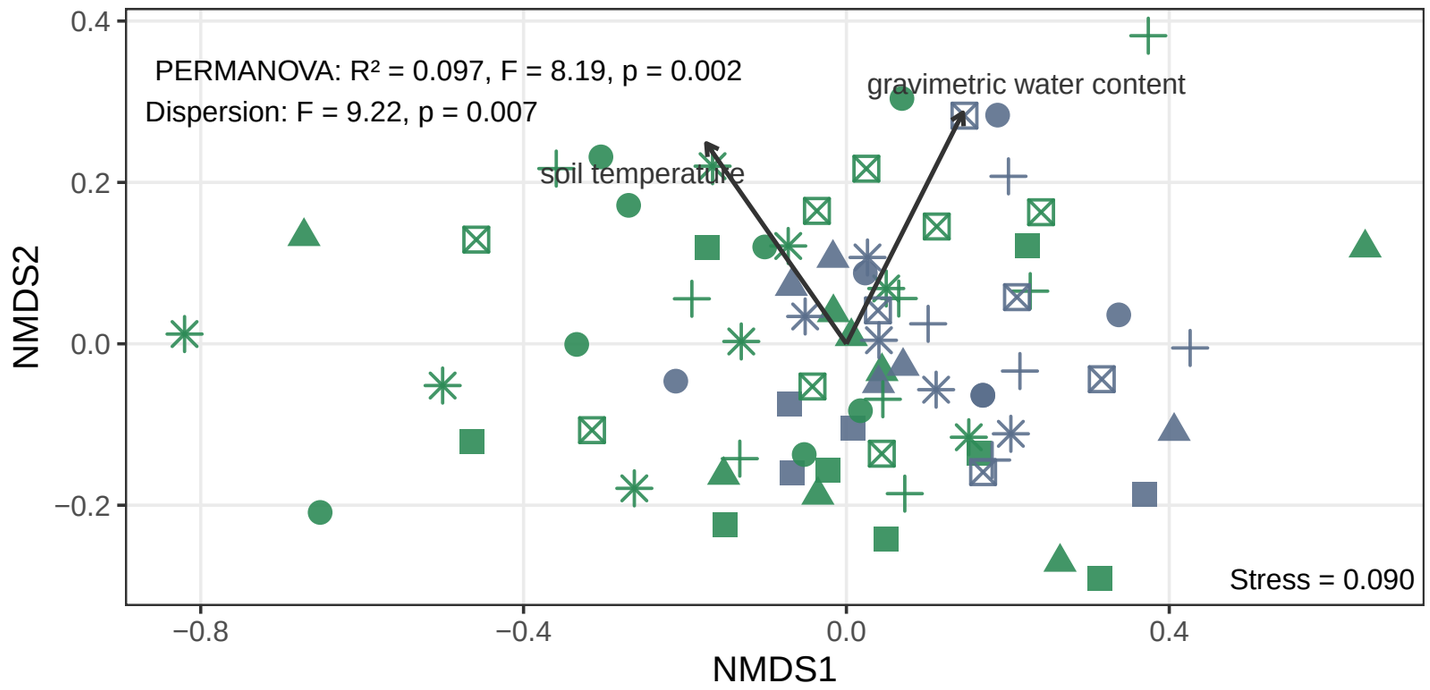
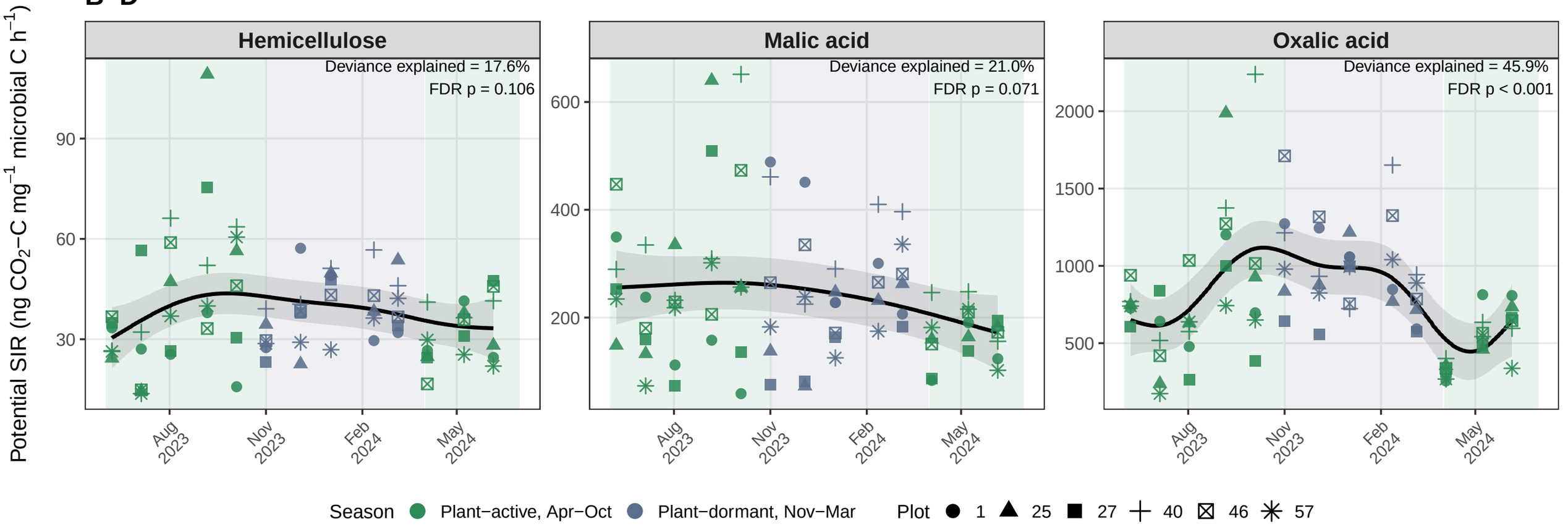
**A**


PERMANOVA: R² = 0.097, F = 8.19, p = 0.002
Dispersion: F = 9.22, p = 0.007
gravimetric water content
soil temperature
Stress = 0.090
NMDS1
NMDS2
0.4
0.2
0.0
−0.2
−0.8
−0.4
0.0
0.4

**B–D**


Potential SIR (ng CO2−C mg−1 microbial C h−1)
Hemicellulose
Deviance explained = 17.6%
FDR p = 0.106
Malic acid
Deviance explained = 21.0%
FDR p = 0.071
Oxalic acid
Deviance explained = 45.9%
FDR p < 0.001
Aug 2023
Nov 2023
Feb 2024
May 2024
Season
Plant−active, Apr−Oct
Plant−dormant, Nov−Mar
Plot
1
25
27
40
46
57